



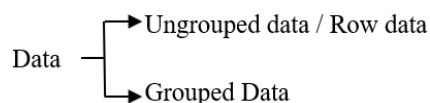
The number of permutations of n objects, where p objects are of one kind, q objects are of another kind and the rest, if any, are of a different kind is $\frac{{}^n P_r}{p!q!}$.

Combination with Repetition

Number of combinations of ' n ' distinct things taking ' r ' at a time when each thing may be repeated any number of times is given by ${}^{n-1+r}C_r$.

3.4 STATISTICS

Statistics → Collection and Analysis of Data



3.4.1 Analysis of Ungrouped Data

If $x_1, x_2, x_3, \dots, x_n$ are ' n ' observations, then

- (1) The range of the data = $R = \max\{x_1, x_2, \dots, x_n\} - \min\{x_1, x_2, x_3, \dots, x_n\}$
- (2) Arithmetic mean : Mean of the data is equal to sum of observations divided by the total number of observations.

$$\bar{x}(\text{or})\mu = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n} = \bar{x} = \mu$$

- The mean of 1st ' n ' natural numbers = $\frac{\frac{n(n+1)}{2}}{n} = \frac{n+1}{2}$
- The mean of 1st ' n ' odd numbers = $\frac{n^2}{n} = n$
- The mean of 1st ' n ' even numbers = $n + 1$

3.4.2 Median

The middle most observation of the data $(x_1, x_2, x_3, \dots, x_n)$ When the data is arranged in either ascending or descending order.

If $x_1, x_2, x_3, x_4, \dots, x_n$ are ' n ' observations that are arranged in ascending/descending order then

- (i) Median of the Data = $\left(\frac{n+1}{2}\right)^{th}$ observation, if ' n ' is odd.
- (ii) Median of the Data = Mean of $\left(\frac{n}{2}\right)^{th}$ & $\left(\frac{n}{2} + 1\right)^{th}$ observations, if ' n ' is even.

3.4.3 Mode

The observation with highest frequency is called mode.

Any Data with two Modes is called → Bimodal Data

If $x_1, x_2, x_3, \dots, x_n$ are ' n ' data points, $\bar{x} = \mu = \frac{x_1 + x_2 + \dots + x_n}{n}$

Mean Deviation of the observation $(x_i) = d_i = x_i - \bar{x}$

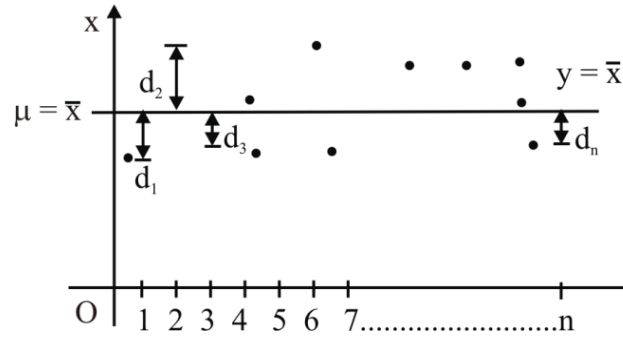


Fig. 5.6. Discrete data

$$\begin{aligned} \text{Sum of derivations of all the observations} &= \Sigma d_i = (x_1 - \bar{x}) + (x_2 - \bar{x}) + \dots + 0 + (x_n - \bar{x}) \\ &= \Sigma d_i = (x_1 + x_2 + \dots + x_n) - n\bar{x} \\ &\boxed{\Sigma d_i = 0} \end{aligned}$$

The sum of mean deviations of all the observations is equal to zero.

3.4.4 Absolute Mean Deviation

If $x_1, x_2, x_3, \dots, x_n$ are 'n' data points with Mean = $\bar{x}$, then the absolute mean deviation of x_i about $\bar{x}$ is given by $|d_i| = |x - \bar{x}|$. The sum of absolute mean derivations of given data is not zero.

$$(\Sigma |d_i| \neq 0) \Rightarrow (|x_1 - \bar{x}| + |x_2 - \bar{x}| + \dots + |x_n - \bar{x}| \neq 0)$$

3.4.5 Standard Deviation

If $x_1, x_2, x_3, \dots, x_n$ ('n' is very large), then the standard deviation of the data is given by

$$\text{Population Standard Deviation } \sigma = \sqrt{\frac{1}{n} \Sigma (x_i - \bar{x})^2}, \quad n \rightarrow \text{size of population}$$

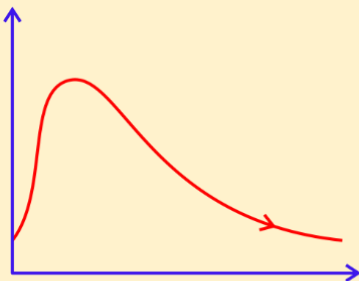
$$\text{Sample Standard derivation: } \sigma = \sqrt{\frac{1}{(n-1)} \Sigma (x_i - \bar{x})^2}, \quad n \rightarrow \text{size of sample}$$

Generally ($n > 29 \rightarrow$ population) ($n < 29 \rightarrow$ sample)

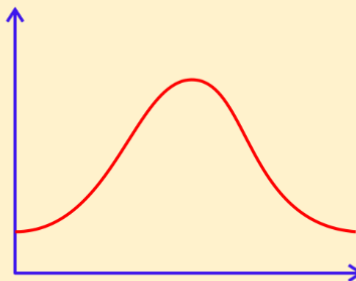
Note: Measures of skewness (The degree of asymmetry)

A lack of symmetry is skewness.

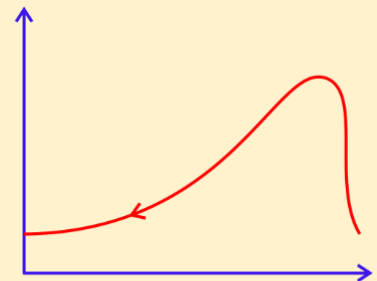
- For symmetric distribution mean (M) = Median (M_d) = Mode (M_e)
- For negatively skewed distribution mean (M) < Median (M_d) < Mode (M_e)
- For positively skewed distribution Mean (M) > Median (M_d) > Mode M_e .



(a) Positively Skewed



(a) Symmetric



(a) Negatively Skewed